

OVERVIEW OF SQL STATEMENTS :

These statements can be divided into 5 Languages ,

1. DATA DEFINITION LANGUAGE (DDL)
2. DATA MANIPULATION LANGUAGE (DML)
3. TRANSACTION CONTROL LANGUAGE (TCL)
4. DATA CONTROL LANGUAGE (DCL)
5. DATA QUERY LANGUAGE (DQL)

1. DATA DEFINITION LANGUAGE (DDL) :

This language is used to construct , modify or remove an object from the DB.

it has 5 statements :

- i.CREATE
- ii.RENAME
- iii.ALTER
- iv.TRUNCATE
- v.DROP

i.CREATE : This statement is used to build/construct an object in the DB.

> Object can be a Table or View (Virtual Table).

1: TO CREATE A TABLE :

Syntax:

CREATE TABLE Table_Name

(
column_name1 datatype constraint_type ,
column_name2 datatype constraint_type ,
column_name3 datatype constraint_type ,

.
.
.
.

```
column_nameN datatype constraint_type  
);
```

Example 1 : to create a PRODUCT table

```
CREATE TABLE PRODUCT  
(  
  PID NUMBER(4) PRIMARY KEY ,  
  PNAME VARCHAR(15) NOT NULL ,  
  PRICE NUMBER(8,2) NOT NULL CHECK( PRICE > 0 )  
);
```

>> TO DESCRIBE THE TABLE:

SYNTAX: DESC TABLE_NAME;

EX: DESC PRODUCT;

TO ASSIGN PRIMARY KEY AND FOREIGN KEY CONNECTION:

SYNTAX:

```
CREATE TABLE Table_Name  
(  
  column_name1 datatype constraint_type ,  
  column_name2 datatype constraint_type ,  
  column_name3 datatype constraint_type ,  
  column_name4 datatype REFERENCES TABLE_NAME(COLUMN_NAME)  
);
```

Example 2 : to create a CUSTOMER table AND connect primary and foreign key.

```
CREATE TABLE CUSTOMER  
(  
  CID NUMBER(3) PRIMARY KEY,  
  CNAME VARCHAR(30) NOT NULL,  
  ADDRESS VARCHAR(30) NOT NULL,  
  PID NUMBER(4) REFERENCES PRODUCT(PID)
```

);

2. TO COPY A TABLE:

SYNTAX: CREATE TABLE TABLE_NAME
AS
SELECT STATEMENT ;

EX: CREATE TABLE SALESMAN
AS
SELECT *
FROM EMP
WHERE JOB = 'SALESMAN' ;

3. TO CREATE A VIEW (VIRTUAL TABLE):

Syntax:

CREATE VIEW View_Name
AS
SELECT statement ;

Example :

CREATE VIEW SALESMAN
AS
SELECT *
FROM EMP
WHERE JOB ='SALESMAN' ;

ii. RENAME : This statement is used to change the name of an existing object .

Syntax:

RENAME Table_Name TO New_Table_Name ;

Example :

RENAME EMP TO EMPLOYEE ;

iii. ALTER : This statement is used to modify the structure of the Object .

> i . TO ADD A COLUMN :

SYNTAX: ALTER TABLE Table_Name

ADD Column_Name Datatype ConstraintType ;

EXAMPLE : ALTER TABLE CUSTOMER
ADD MAIL_ID VARCHAR(25) UNIQUE NOT NULL ;

> ii. TO DROP A COLUMN :

SYNTAX: ALTER TABLE Table_Name
DROP COLUMN Column_Name ;

EXAMPLE : ALTER TABLE CUSTOMER
DROP COLUMN MAIL_ID ;

> iii. TO RENAME A COLUMN :

SYNTAX: ALTER TABLE Table_Name
RENAME COLUMN Column_Name TO New_Column_Name ;

EXAMPLE : ALTER TABLE CUSTOMER
RENAME COLUMN CNO TO CONTACT_NO ;

> iv. TO MODIFY THE DATATYPE :

SYNTAX: ALTER TABLE Table_Name
MODIFY Column_Name New_Datatype ;

EXAMPLE : ALTER TABLE CUSTOMER
MODIFY ADDRESS CHAR(25) ;

> v. TO MODIFY THE CONSTRAINT :

SYNTAX: ALTER TABLE Table_Name
MODIFY Column_Name Existing_Datatype CONSTRAINT ;

EXAMPLE : ALTER TABLE CUSTOMER
MODIFY ADDRESS CHAR(25) NOT NULL ;

4. TRUNCATE : This statement is used to remove all the records from the table permanently . (it will not destroy the table structure)

SYNTAX: TRUNACTE TABLE Table_Name ;

Example : TRUNCATE TABLE SALESMAN ;

5. DROP : This statement is used to remove the object from the DB.

SYNTAX: DROP TABLE Table_Name ;

EXAMPLE : DROP TABLE SALESMAN ;

> TO RECOVER THE TABLE FROM RECYCLE BIN:

SYNTAX: FLASHBACK TABLE Table_Name
TO BEFORE DROP ;

EXAMPLE : FLASHBACK TABLE SALESMAN
TO BEFORE DROP ;

> TO REMOVE THE TABLE FROM RECYCLE BIN :

SYNTAX: PURGE TABLE Table_Name ;

EXAMPLE : PURGE TABLE SALESMAN ;

NOTE : These statements are auto-commit statements. (Automatically gets saved in the DB)

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DATA MANIPULATION LANGUAGE :-

This language is used to manipulate the object by performing insertion , updation and deletion.

it has 3 statements :

1. INSERT
2. UPDATE
3. DELETE

1. INSERT : This statement is used to add records into the table .

Syntax: INSERT INTO Table_Name VALUES(v1 , v2 , , Vn);

Example : INSERT INTO PRODUCT VALUES(1 , 'iPhone 12' , 35000 , null);

Syntax : INSERT INTO Table_Name VALUES(&c1 , &c2 , &c3 , &c4);
enter the value for c1 :
enter the value for c2 :
enter the value for c3 :
enter the value for c4 :

- (to re-execute use /)

Example : INSERT INTO customer VALUES(&cid , &cname , &contact_no , &address , &pid);
enter the value for cid : 111
enter the value for cname: 'RAMA'
enter the value for address: 'QSP CHROMPET'
enter the value for pid : 2

2. UPDATE : This statement is used to modify the exsisting values in the table .

Syntax: UPDATE Table_Name
SET Column_Name = Value , Column_Name = Value
[WHERE <filter_condition>]

Example : UPDATE PRODUCT
SET DISCOUNT = PRICE*3/100 ;

Example : UPDATE PRODUCT
SET DISCOUNT = PRICE*5/100
WHERE PRICE > 100000 ;

3. DELETE : This statement is used to remove a particular record from the table .

Syntax: DELETE FROM Table_Name
[WHERE <filter_condition>] ;

Example : DELETE FROM AMAZON_SHOP
WHERE PRICE > 100000 ;

Example : DELETE FROM EMP
WHERE JOB ='CLERK' ;

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iii. TRANSCATION CONTROL LANGUAGE :

This language is used to control the flow of transactions on the DB .
It has 3 statements :

1. COMMIT : This statement is used to save the transcatons in to the DB.

Syntax: COMMIT ;

2. ROLLBACK :This statement is used to GO BACK to the latest comitted point on the DB.

Syntax: ROLLBACK ;

EX : ROLLBACK TO SAVEPOINT_NAME ;

3. SAVEPOINT : This statement is used to mark the positions on the DB.

Syntax: SAVEPOINT Savepoint_Name ;

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iv. DATA CONTROL LANGUAGE :

This language is used to control the flow of data between the users .
it has 2 statements :

1. GRANT : This statement is used to give permission to a user .

Syntax: GRANT sql_statement
ON Table_Name
TO User_Name ;

Example : GRANT SELECT
ON PRODUCT
TO HR ;

2. REVOKE : This statement is used to take back the permission from the user .

Syntax: REVOKE sql_statement
ON Table_Name
FROM User_Name ;

Example : REVOKE SELECT
ON PRODUCT
FROM HR ;

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QUESTIONS ON STATEMENTS:

1. WAQ TO UPDATE SALARY OF SMITH TO 2000

2. WAQ TO UPDATE SALARY OF KING TO TWICE OF HIS SALARY
3. WAQ TO CONSTRUCT A TABLE(STUDENT) THAT CAN ACCEPT
 - a. STUDENT_ID
 - b. STUDENT_NAME
 - c. BRANCH
 - d. PERCENTAGE
 - e. NO_of_Backlogs
4. WAQ TO DELETE THE DETAILS OF EMPLOYEE'S WHO'S NAME STARTS WITH A CHARACTER 'T'
5. WAQ TO REMOVE ADAMS FROM EMP TABLE
6. WAQ TO UPDATE SALARIES OF THE EMPLOYEES TO 3550 WHO ARE FROM NEW YORK
7. WAQ TO CREATE TABLE NAMED LIBRARY WITH
 - a. BOOK_NO
 - b. BOOK_NAME
 - c. AUTHOR
 - d. PUBLISHED ON
 - e. STUDENT_ID(FROM STUDENT TABLE)
8. WAQ TO ADD ATLEST 6 RECORDS INTO LIBRARY TABLE .
9. WAQ TO ESTABLISH A CONNECTION BETWEEN STUDENT AND LIBRARY TABLE
- 10.WAQ TO ADD RECORDS TO STUDENT TABLE (ATLEST 6).
11. WAQ TO CHANGE COMMISSIONS OF EMPLOYEES TO 300 ONLY IF THEY DONT EARN COMMISSION
12. WAQ TO UPDATE SALARY OF EMPLOYEES WORKING IN CHICAGO WITH 20% HIKE IN THE SALARY .
13. WAQ TO DELETE THE EMPLOYEES WHO ARE HIRED BEFORE THE YEAR 81.
14. WAQ TO ADD 2 RECORDS INTO THE EXISTING EMPLOYEE TABLE AND THEY MUST WORK IN DEPT 40 .
15. WAQ TO GIVE A WELCOME BONUS OF RS 1000 FOR THE EMPLOYEES OF DEPT 40 (SAL+1000)

